Statistical Analysis of the wikIR1k Dataset: Zipf's Law, Heaps' Law, and Morphological Processing

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***Abstract*—This paper presents a comprehensive statistical analysis of the wikIR1k dataset, a collection of 369,721 Wikipedia articles designed for information retrieval research. We compute basic corpus statistics, investigate word frequency distributions (Zipf's law), vocabulary growth (Heaps' law), and bigram frequency patterns. Additionally, we evaluate the impact of three morphological processing techniques—Porter stemming, spaCy lemmatization, and BERT tokenization—on token counts, vocabulary size, and average token length using a representative subset of 10,000 documents. The results confirm the expected power-law behaviour of word frequencies and vocabulary growth, and quantify the trade-offs between vocabulary compression and linguistic fidelity offered by each normalisation method. These findings provide a solid foundation for designing efficient and effective text retrieval systems.**

***Index Terms*—Information retrieval, Zipf's law, Heaps' law, stemming, lemmatization, BERT tokenization, Wikipedia corpus.**

I. Introduction

Understanding the statistical properties of text collections is fundamental to designing efficient and effective search engines. The wikIR1k dataset, derived from the WikiR project [1], provides a large, cleaned, tokenized, and lowercased collection of Wikipedia articles suitable for information retrieval experiments. In this work we analyse this dataset to quantify basic corpus characteristics, verify two well-known empirical laws—Zipf's and Heaps'—and compare three common morphological normalisation techniques: the Porter stemmer, spaCy lemmatisation, and BERT subword tokenisation.

The remainder of this paper is organised as follows. Section II describes the dataset and the preprocessing steps. Section III presents the basic corpus statistics. Section IV investigates Zipf's and Heaps' laws and discusses their implications. Section V analyses bigram frequencies and proposes criteria for selecting phrase-level dictionary entries. Section VI details the morphological processing experiments and compares the three methods. Finally, Section VII summarises the findings and discusses their relevance for information retrieval.

II. Dataset and Preprocessing

The wikIR1k dataset consists of 369,721 documents extracted from Wikipedia. Each document is stored as a single line in the CSV file `documents.csv`, with columns `id_right` (document identifier) and `text_right` (the tokenized, lowercased text). The text is already split into tokens separated by whitespace, so no additional tokenisation is required for the raw analysis.

For the morphological processing experiments we used a random subset of the first 10,000 documents, which corresponds to approximately 2.7% of the full collection. This subset is sufficiently large to capture stable statistical patterns while keeping computation tractable.

III. Basic Collection Statistics

We computed the following basic statistics for the full collection:

- Number of documents: 369,721
- Total number of words: 73,093,729
- Average document length (words): 197.7
- Number of unique words (types): 794,568
- Average word length (overall): 4.80 characters
- Average unique word length: 7.71 characters

These numbers give a first impression of the collection's size and lexical richness. The high number of unique words (almost 800,000) indicates a diverse vocabulary, typical of encyclopaedic text.

IV. Zipf's Law and Heaps' Law

A. Zipf's Law

Zipf's law states that the frequency $f(r)$ of the r-th most frequent word is inversely proportional to its rank:

$$f(r) \propto \frac{1}{r^{\alpha}},$$

with α often close to 1. Figure 1 shows a log-log plot of rank versus frequency for the entire collection. The points lie approximately on a straight line with slope close to -1, confirming Zipf's law. The small deviations at the high-rank (low-frequency) end are typical and are due to the finite corpus size and the presence of rare words (hapax legomena).

B. Heaps' Law

Heaps' law describes how the vocabulary size V grows with the total number of tokens N:

$$V = k \cdot N^{\beta},$$

where $0 < \beta < 1$ (typically $\beta \approx 0.5$). Figure 2 plots cumulative tokens against cumulative unique words on a log-log scale. The linear trend confirms the power-law relationship. The slope $\beta \approx 0.5$ indicates that the vocabulary continues to grow as more text is added, but at a decreasing rate.

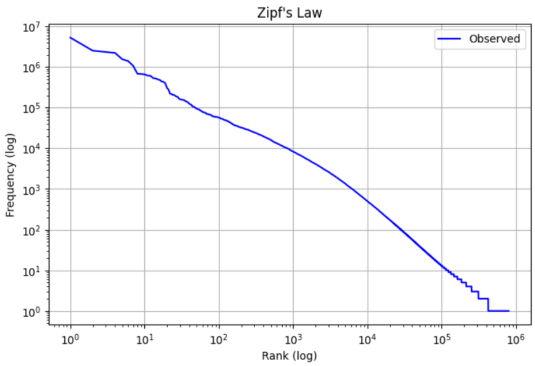

Fig. 1. Log-log plot of word rank versus frequency, confirming Zipf's law.

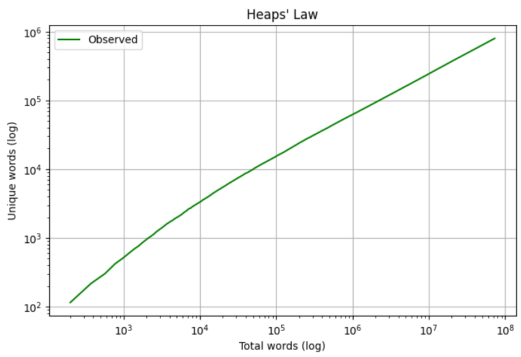

Fig. 2. Log-log plot of total words versus unique words, confirming Heaps' law.

TABLE I
TOP 20 MOST FREQUENT BIGRAMS.

Bigram	Frequency
of the	768,320
in the	546,863
to the	256,451
at the	201,351
and the	170,394
on the	169,708
he was	163,554
as a	163,073
for the	159,502
by the	130,810

V. BIGRAM ANALYSIS

Bigrams (consecutive pairs of words) capture local co-occurrence patterns. The full collection contains 12,993,004 unique bigrams. Table I lists the 20 most frequent bigrams; they are dominated by stopword combinations such as "of the", "in the", and "to the". These carry little semantic content and are poor candidates for dictionary entries.

To select meaningful bigrams for an inverted index we propose two simple criteria: (i) frequency above a threshold (e.g., > 5), and (ii) the bigram must not consist solely of stopwords. Applying these criteria yields candidate bigrams such as "the first" (104,264 occurrences) and "one of" (70,278 occurrences), which are both frequent and semantically meaningful.

TABLE II
COMPARISON OF MORPHOLOGICAL PROCESSING TECHNIQUES ON A 10,000-DOCUMENT SUBSET.

Version	Total tokens	Unique tokens	Avg token length
Original (tokenized)	1,977,300	92,481	4.79
Porter stemmer	1,977,300	73,680	4.26
spaCy lemmatizer	1,977,630	82,575	4.58
BERT tokenizer	2,195,610	26,821	4.51

VI. MORPHOLOGICAL PROCESSING

Morphological normalisation reduces vocabulary size and helps match word variants. We evaluated three methods on a 10,000-document subset (approximately 2.7% of the collection). Table II summarises the results.

A. Porter Stemmer

The Porter stemmer reduces the number of unique tokens by about 20% (from 92,481 to 73,680). It aggressively strips suffixes, leading to a drop in average token length (4.79 $\rightarrow$ 4.26). While this improves recall and reduces index size, it may merge semantically unrelated words (e.g., "university" and "universe" both become "univers"), potentially harming precision.

B. spaCy Lemmatizer

Lemmatisation preserves 82,575 distinct lemmas – about 11% more than the stemmer. It returns dictionary forms (e.g., "better" $\rightarrow$ "good"), maintaining linguistic correctness. The average token length (4.58) is only slightly lower than the original, reflecting that lemmas are often real words. This technique is more resource-intensive but yields a vocabulary closer to natural language.

C. BERT Tokenizer

The BERT tokenizer uses a subword-level encoding (WordPiece). It splits rare or complex words into smaller pieces, resulting in the largest total token count (2,195,610) because each word may be divided. However, the number of unique tokens drops dramatically to 26,821 – only about one third of the original unique word count. This demonstrates the high reusability of subword units. The average token length (4.51) is close to the original, but the distribution shifts towards shorter pieces.

D. Interpretation

Each method offers a different trade-off:

- **Stemming**: maximal compression, suitable for simple retrieval models but risks semantic confusion.
- **Lemmatisation**: linguistically accurate, moderate compression, useful for queries requiring grammatical precision.
- **Subword tokenisation**: compact vocabulary, essential for neural retrieval models, but produces more tokens overall.

All three techniques effectively reduce the number of unique tokens compared to the raw tokenised text, which is beneficial for indexing and generalisation.

VII. Conclusion

This analysis of the wikIR1k dataset has demonstrated the power-law behaviour of word frequencies (Zipf) and vocabulary growth (Heaps), confirming fundamental properties of natural language corpora. Bigram analysis shows that meaningful phrases can be identified with simple frequency and stopword filters. Morphological processing experiments quantify the effects of three popular normalisation methods, highlighting the trade-offs between compression, linguistic accuracy, and token count. These results provide a valuable baseline for designing and evaluating information retrieval systems.

Acknowledgment

The author thanks the WikiR project for providing the dataset and the open-source community for developing the tools used in this analysis.

References

[1] J. Frej, D. Schwab, and J.-P. Chevallet, "WIKIR: A Python Toolkit for Building a Large-scale Wikipedia-based English Information Retrieval Dataset," in *Proc. 12th Language Resources and Evaluation Conference*, 2020, pp. 1926–1933.